

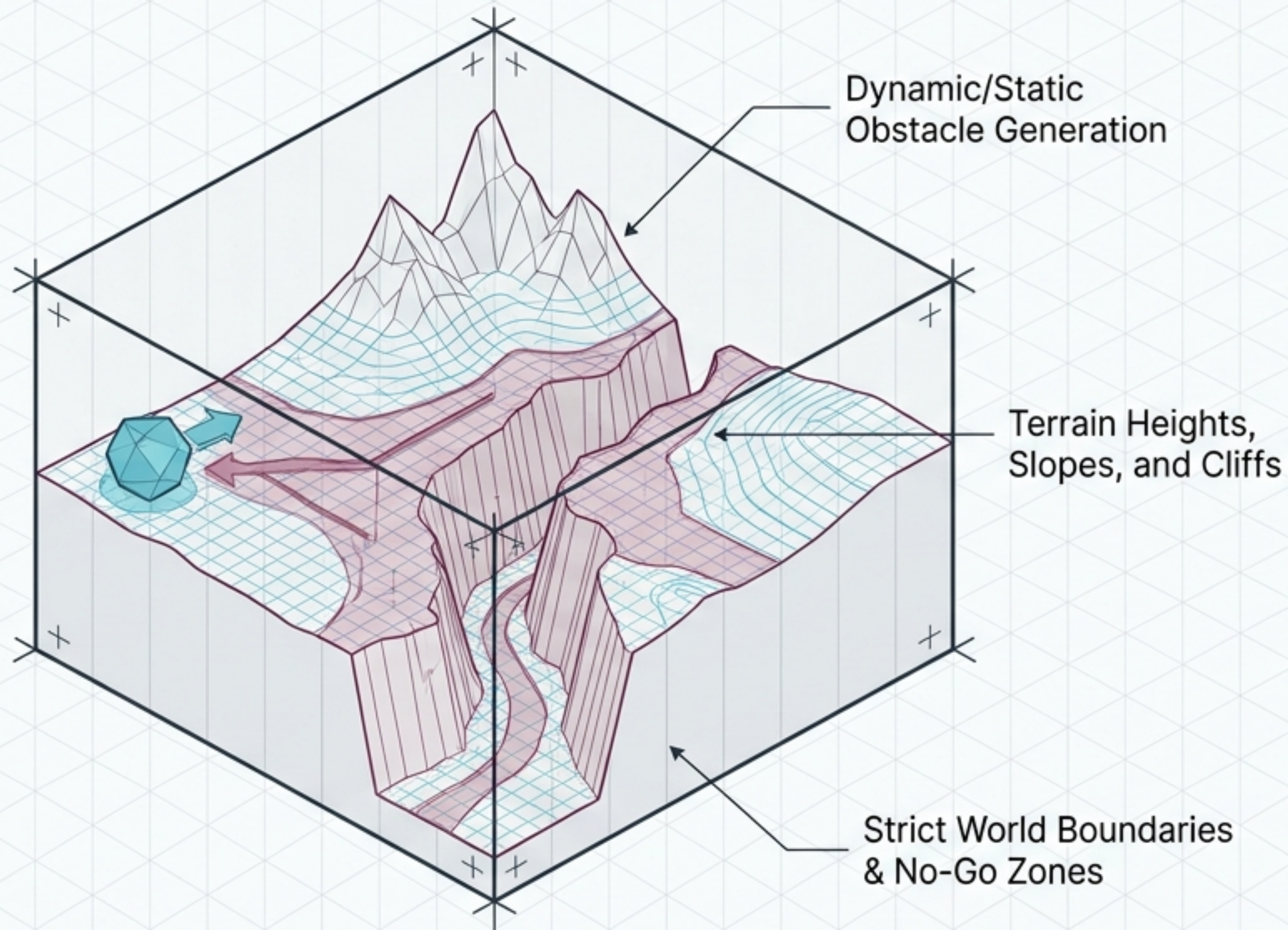
Batch: 2023–27

INTELLIGENT DRONE ROUTE PLANNING ALGORITHM

Software Simulation & Architecture Blueprint

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Architecting the Simulated Environment



The Drone State Engine

Physics Tracking:

Position, Velocity, Acceleration, 3D Orientation (Angular Velocity), Altitude.

POS: X 12.4, Y -5.1, Z 30.2 m

VEL: 8.5 m/s

ACC: 0.3 g

ORIENT: P 5°, R 2°, Y 180°

ALT: 30.2 m

Operational State:

Real-time Battery Depletion



BATT: 78%

Flight Validity Checks



VALID

Active Mission State

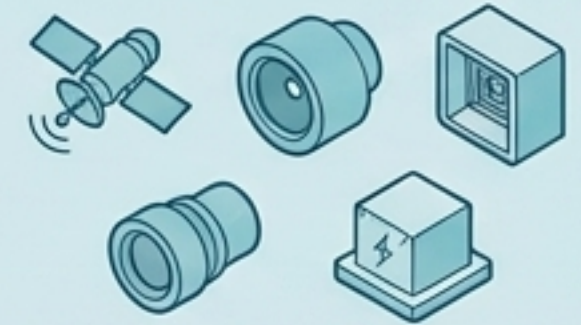
ACTIVE: WAYPOINT 3

The Intelligence Core: Perception to Trajectory

Layer 1: The WHY (Information & Perception)

Simulated IMU, GPS, Altimeter, Depth, LiDAR, and RGB Camera.

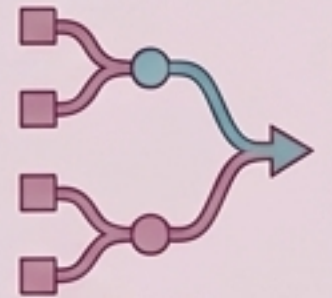
Realism Injectors: Sensor noise, missing measurements, and variable confidence values.



Layer 2: The HOW (Decision Engine)

Path Cost Function: Balancing distance, terrain risk, obstacle proximity, energy limits, and uncertainty.

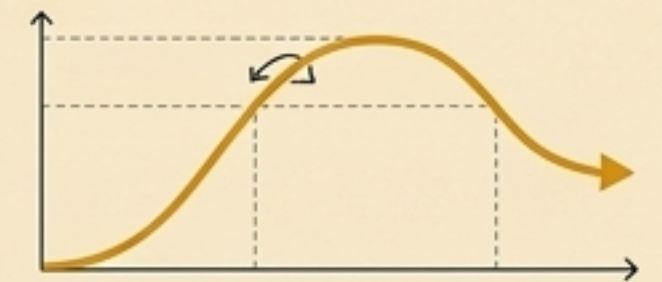
Core Algorithms: A* routing, dynamic local replanning, and emergency fallback.



Layer 3: The WHAT (Trajectory Constraints)

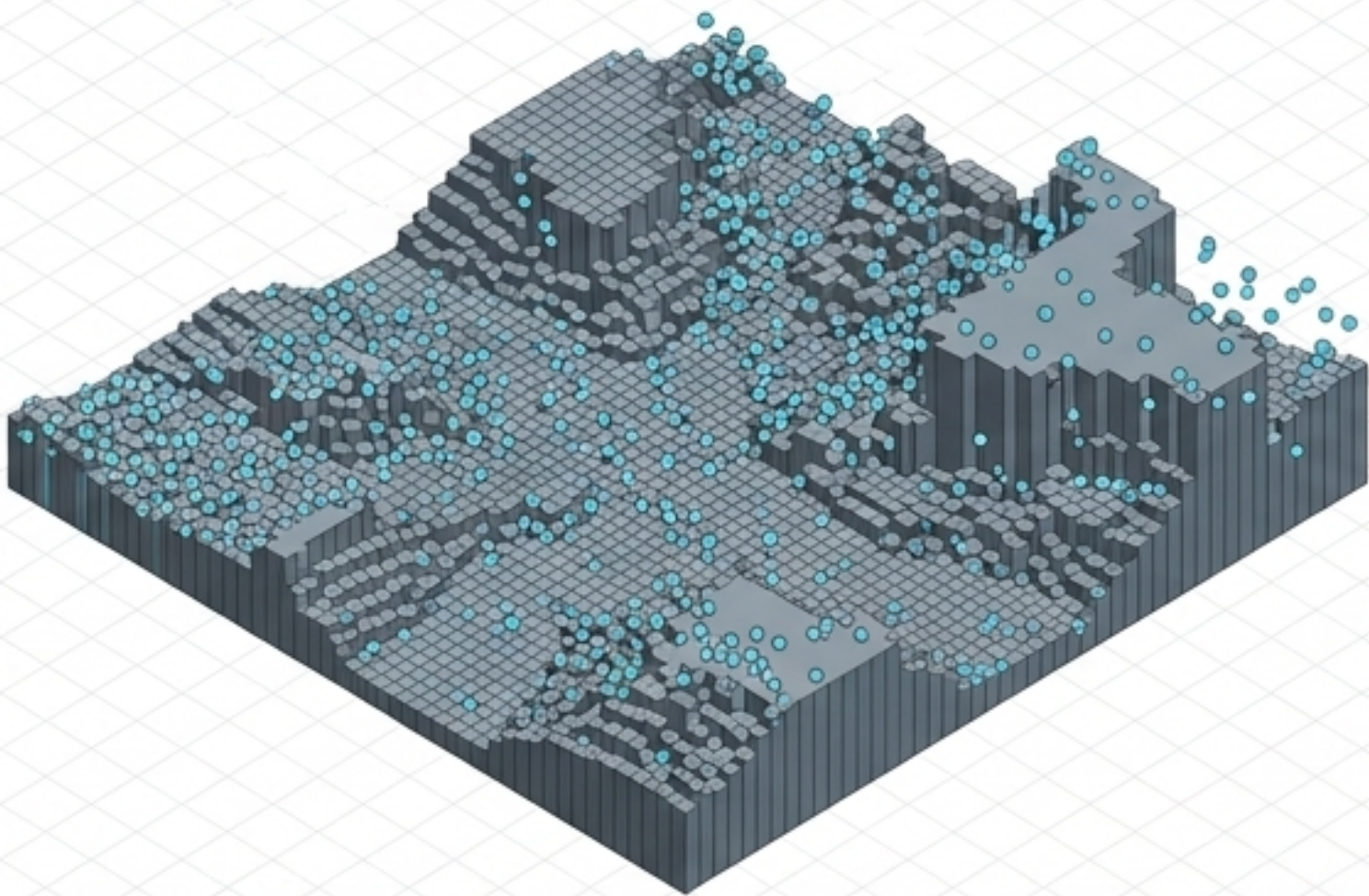
Physically constrained outputs: Curves, climbs, descents, and hover/hold modes.

Strict enforcement of minimum-clearance and velocity/acceleration limits.

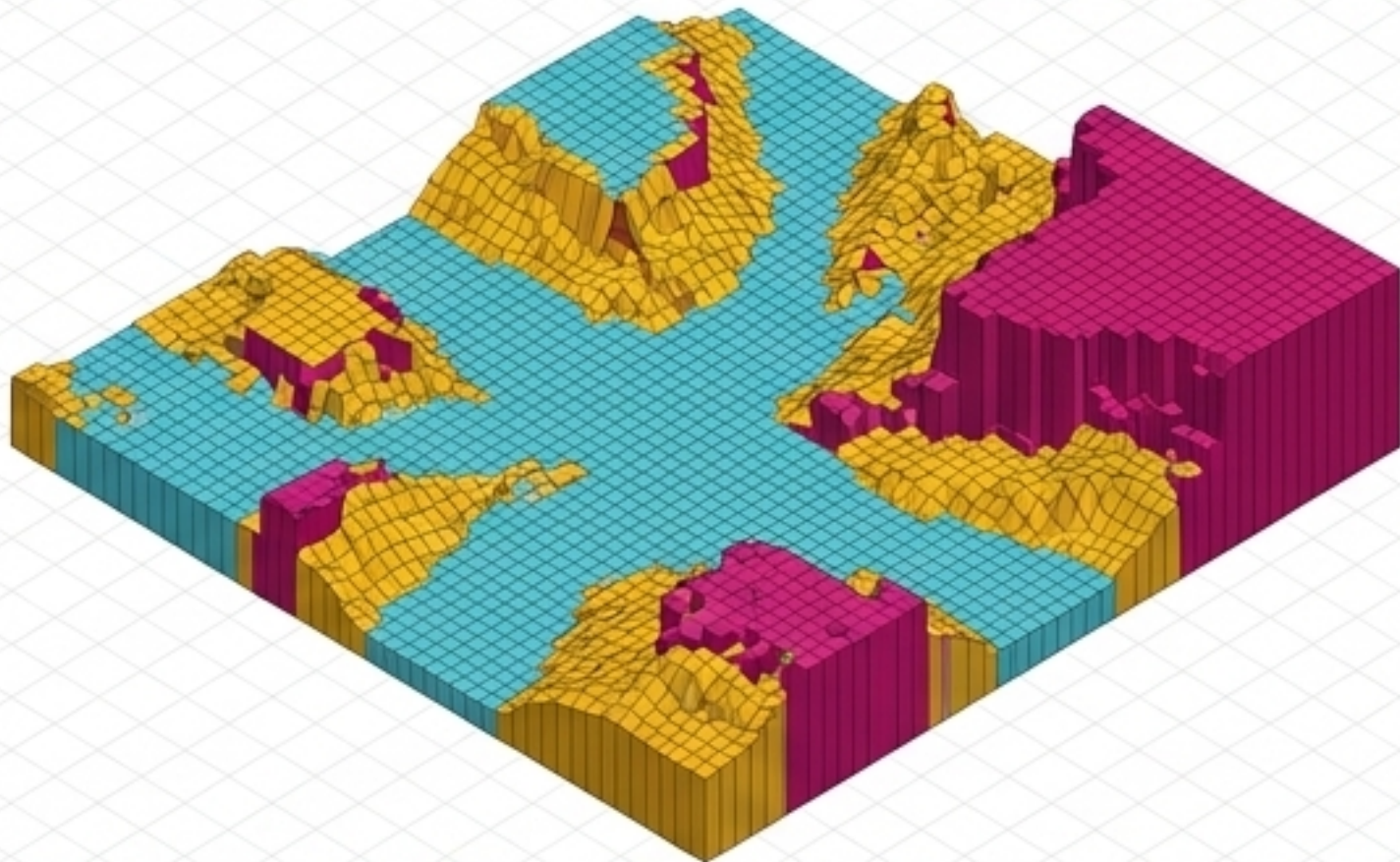


Mapping the Unknown: Traversability & Risk

Raw Input



Processed Output



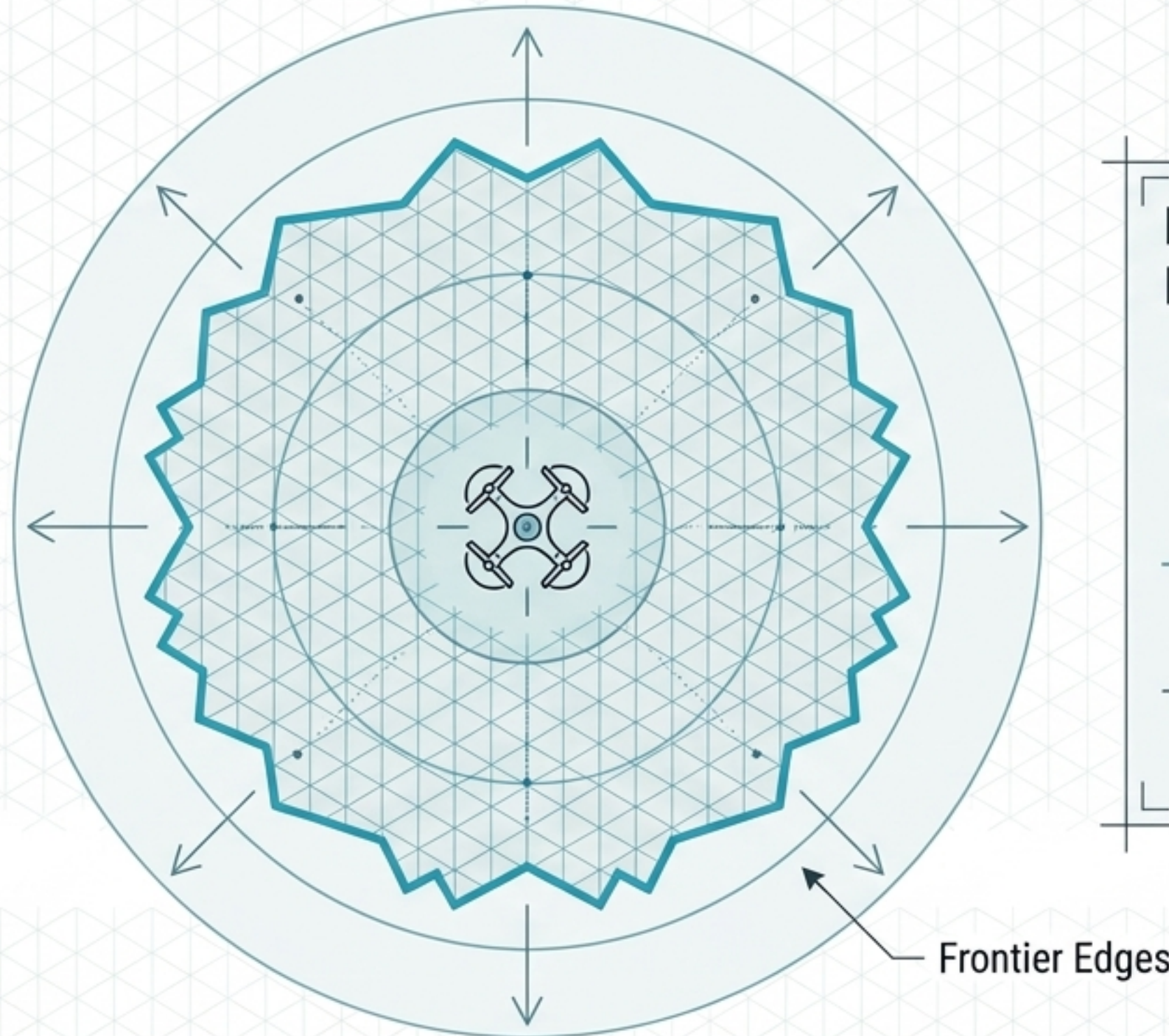
Terrain Cost Calculation Matrix		
1	2	3
Slope Cost	Roughness Cost	Obstacle Cost
Penalizing steep ascents	Avoiding unstable terrain/vegetation	Maintaining buffer zones from static/dynamic threats

Output: Generation of an Occupancy Grid and real-time Local Safe Corridors via simulated sensor observations.

Autonomous Exploration & Rescue Logic

Frontier-Based Exploration

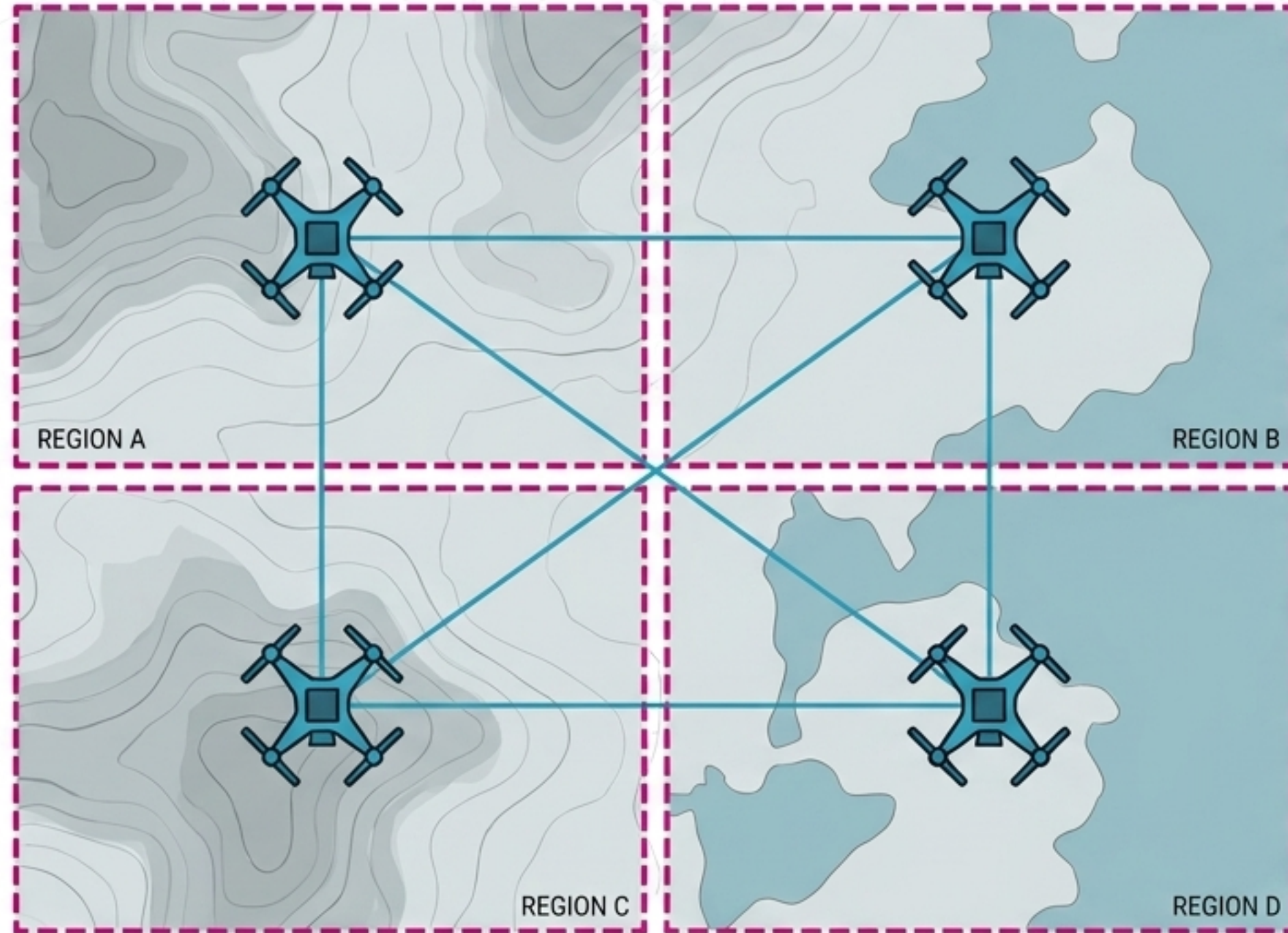
-  **Information-Gain Scoring:** Prioritizing paths that map the highest volume of unknown territory.
-  **Revisit Logic:** Guaranteeing 100% area coverage without dead-locks.
-  **Continuous Re-planning:** Updating routes dynamically as new frontiers are exposed.



Rescue & Target Investigation

-  **Simulated survivor detection** with confidence thresholding.
-  **Immediate** behavioral shift to priority investigation paths.
-  **Automated inspect/hover** maneuvers upon target confirmation.

Swarm Intelligence: Decentralized Coverage



Decentralized Communication

P2P Broadcasts: Position, velocity, battery state, and hazard flags.

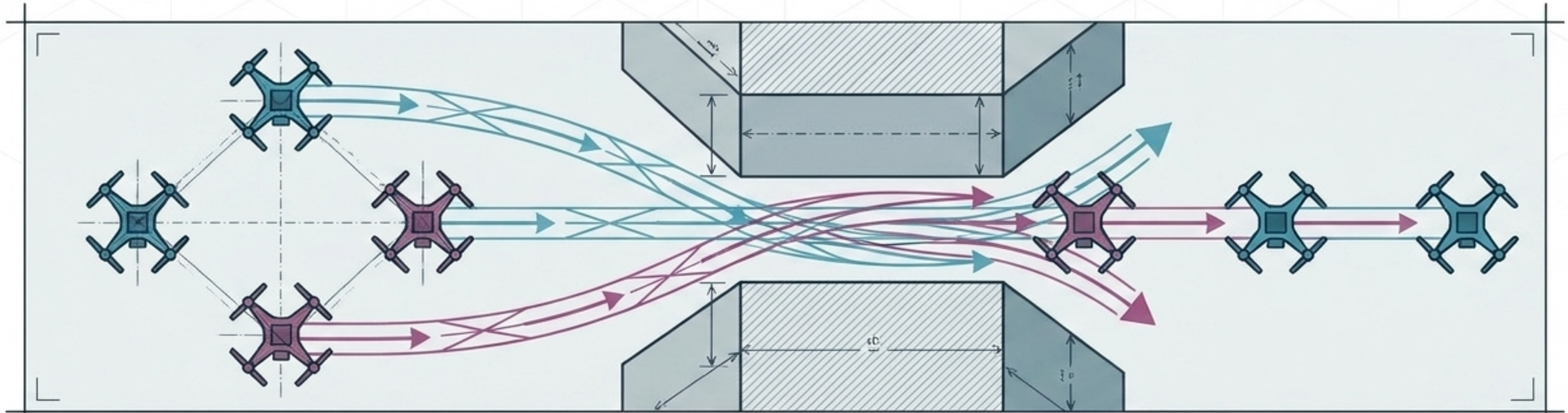
Shared Map Synchronization: No single central controller; drones operate via local autonomy.

Greedy Task Allocation

Dynamic division of the exploration area.

Reassigning blocked regions or taking over coverage ownership if a neighboring drone experiences battery drop or system failure.

Formation Flight & Dynamic Adaptation



Inter-Drone Safety

Pairwise collision detection and strict inter-drone safety distance enforcement.

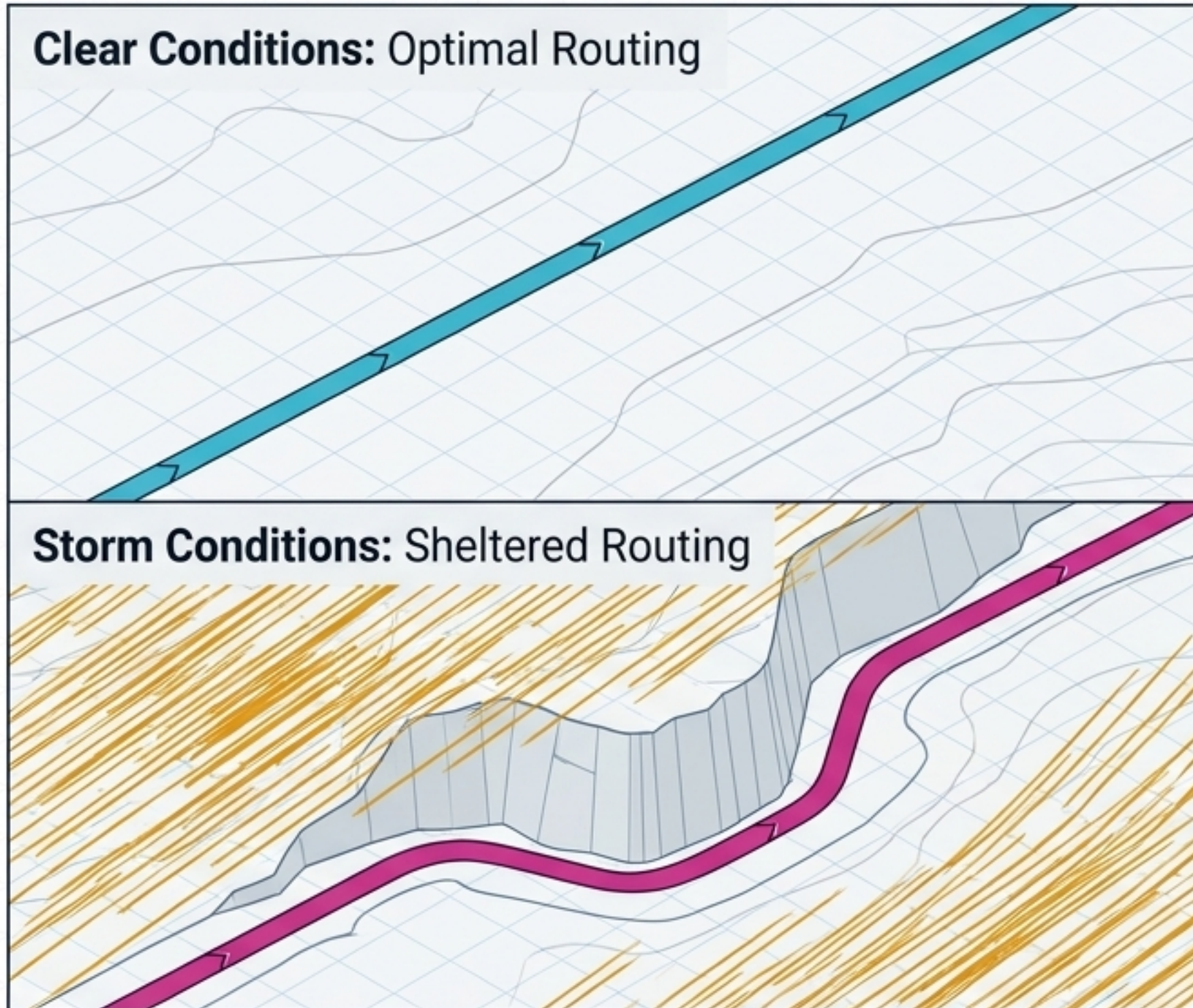
Emergency separation protocols to prevent cascade collisions.

Terrain-Triggered Morphing

Available Formations: Line, Arc, Curve, Wedge, and Diamond.

Adaptive Shifting: Automatically switching formations based on terrain constraints and obstacle proximity.

Stress Testing: Storm & Terrain Resilience

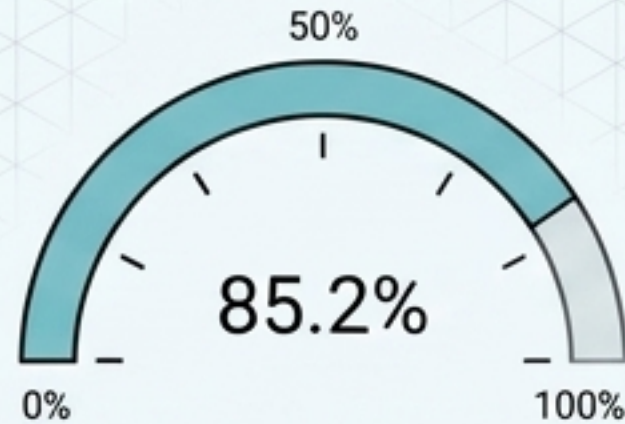


Dynamic Resilience Actions	
Environmental Hazards Simulated:	High wind gusts, dust/smoke (visibility degradation).
Sheltered Routing:	Penalizing exposed regions and actively seeking terrain cover.
Kinematic Throttle:	Dynamic speed reduction in low-visibility zones.
Formation Contraction:	Tightening swarm spacing under high external risk.
Emergency Hold:	Safe-zone return logic during critical weather events.

Diagnostic Matrix: Failure & Fallback Protocols

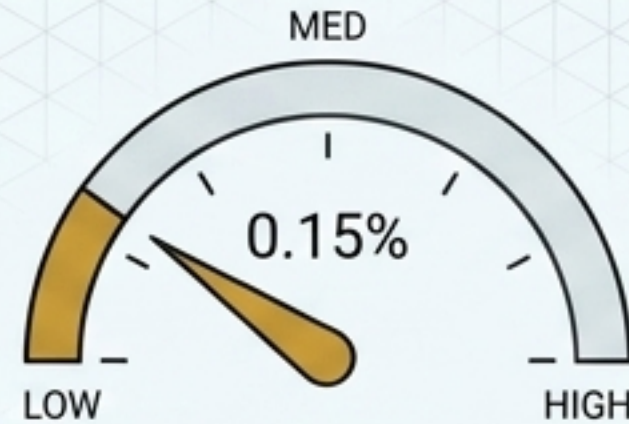
Catastrophic System Failure	Autonomous Fallback Response
GPS Loss	Triggers switch to local mapping & IMU dead reckoning.
Communication Outage	Triggers isolated-drone autonomy & local rejoin logic.
Leader Failure	Swarm dynamically reassigns coverage and elects a new priority node.
Sensor Failure (LiDAR/RGB)	Graceful degradation, relying on secondary sensors (depth/altimeter).
Sudden Path Blockage	Immediate local replanning and emergency hold if no preferred path exists.

Validation Metrics & Software Roadmap



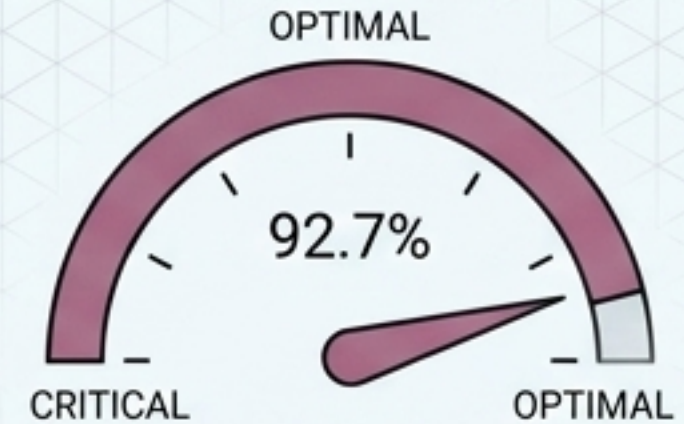
Exploration Coverage %

Measuring total mapped volume and target discovery.



Collision Rate

Tracking pairwise and obstacle clearance success.



Energy Efficiency

Validating route success against simulated battery constraints.



Foundation

Physics, Environment, Drone State

Single Autonomy

Perception, Planning, Exploration

Swarm Capability

Decentralization, Formation, Task Allocation

Stress Validation

Weather Resilience, Failure Fallbacks, Metrics